



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. CSE 'A' & 'B'

Course Code & Title: CCS342 DevOps

Name of the Faculty member (s): Dr.B.Vijayalakshmi, ASP-II/CSE

Innovative Practice Description

- **Unit / Topic: Unit II / Maven Plugins**

- **Course Outcome: CO2**

- **Topic Learning Outcome: 2b**

- **Activity Chosen: Problem based Learning – Treasure hunt cards**

- **Justification:**

There are many maven plugins are available. The Treasure Hunt Card activity transforms the typically theoretical topic of Maven Plugins into an interactive, discovery-based challenge. Instead of passively memorizing plugin names like compiler, surefire, clean, shade, site, dependency, students decode clues, search documentation, and collect plugin facts such as abilities, purposes, and configuration use-cases. In this **Problem-Based Learning (PBL)** approach, students solve micro-problems (e.g., “Which plugin helps package a fat JAR?”), fostering inquiry and active engagement.

- **Time Allotted for the Activity: 30 Minutes**

- **Details of the Implementation:**

- The faculty member designed Treasure Hunt Cards containing real-world problem scenarios related to Maven plugins. Each card included the following:
 - A clue describing a practical development situation.
 - A hint pointing to the relevant plugin (e.g., compiler, surefire, shade, install).
 - A task requiring students to identify the plugin, its purpose, and configuration use.
- The students were divided into 5 small teams of 3–5 members. Each team received a **different application scenario** (e.g., Online Bookstore, E-Commerce App, Library Management System) with its own set of Treasure Hunt Cards. The sample cards were shown in Fig.1. Teams were instructed to **decode clues, research plugins, and record answers**.
- The team members were worked collaboratively to solve the clues by:
 - Identifying the appropriate Maven plugin.
 - Understanding its purpose and configuration
 - Applying the plugin to the example scenario
- Students referred to official Maven documentation, tutorials, and online resources for accurate answers and it is shown in Fig 2.

- Teams **share their findings** with the class, explaining how they solved each clue.
- The instructor clarifies any **misconceptions** and highlights **key functionalities of each plugin**.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	3	2	3	3	2	1	1	2	3	1	1	3	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO2	PO4	PO5	PO9	PO10	PSO2
	2	2	2	1	1	3
Justification for correlation	Students decode clues and identify the correct Maven plugin for a given problem.	Students explore Maven documentation and online resources to solve micro-problems.	Students gain hands-on familiarity with Maven plugins	Teams collaborate to solve clues, discuss answers, and cross-verify results.	Teams present their findings, explaining the plugin functionality to peers..	The idea of different devops cloud tools are essential to developing IT solutions.

• **Images / Screenshot of the practice:**

Team 1 – Online Bookstore Application

Clue 1: Your code speaks Java 11, but Maven must understand it too. Which helper will make sure it compiles the right language? <i>Hint:</i> Think compiler.
Clue 2: Before a book is printed, proofreaders check for mistakes. Which Maven helper plays the role of the tester? <i>Hint:</i> Think about running JUnit tests.
Clue 3: Customers don't want scattered pages; they want one complete book. Which plugin binds all dependencies into one executable JAR? <i>Hint:</i> Shade or Assembly.
Clue 4: The book is ready, now place it safely in the library shelf (local repo) so others can borrow it. Which plugin installs it? <i>Hint:</i> Think install.

Team 2 – Student Attendance Management System

Clue 1: Teachers need a report card. Which plugin writes HTML documentation for your project? <i>Hint:</i> Think 'site'.
Clue 2: The class monitor checks if students wear uniforms properly. Which plugin checks if your code follows rules? <i>Hint:</i> Style checker.
Clue 3: Final exams happen after midterms. Which plugin runs integration tests after packaging? <i>Hint:</i> It is a 'failsafe' way.
Clue 4: Once attendance is recorded, results are uploaded to the server. Which plugin helps deploy your app to Tomcat? <i>Hint:</i> Tomcat-specific.

Team 3 – Weather Forecasting REST API

Clue 1: You want your Spring Boot service to run without typing java -jar. Which plugin lets you run the app directly with Maven? <i>Hint:</i> Spring Boot.
Clue 2: Forecasts need accuracy; wrong predictions cause trouble. Which plugin scans your code for bugs and design flaws? <i>Hint:</i> PMD or SpotBugs.
Clue 3: You want to put your weather station inside a shipping container (Docker). Which plugin builds Docker images? <i>Hint:</i> Spotify or Jib.
Clue 4: The forecast data should be shared with everyone, not just local users. Which plugin publishes artifacts to a remote repo? <i>Hint:</i> Deploy.

Team 4 – E-Learning Portal

Clue 1: Students need both lessons and notes. Which plugin generates Javadocs for your project? <i>Hint:</i> Javadoc.
Clue 2: After exams, results are displayed as reports. Which plugin generates HTML reports for unit test results? <i>Hint:</i> Surefire report.
Clue 3: Hackers may slip viruses into downloaded dependencies. Which plugin scans for vulnerabilities? <i>Hint:</i> OWASP.
Clue 4: An online portal must be a website, not just files. Which plugin packages the project as a WAR file? <i>Hint:</i> WAR.

Fig. 1: Sample treasure Hunt Cards



Fig 2. Discussion Session

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Many appreciated the interactive and gamified approach, which made learning Maven plugins easier to understand and remember.
 - Students found the activity **highly engaging** compared to traditional lectures, as it turned theoretical concepts into a hands-on challenge.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Encouraged active, self-directed learning, leading to better retention than conventional memorization.
 - Promoted critical thinking and problem-solving through micro-problems like “Which plugin helps package a fat JAR?”

- ❖ ***Challenges faced in implementation:***

- Some students initially struggled to interpret clues or locate the correct documentation.
 - Time management was a challenge, as teams spent differing amounts of time on each card.

References:

- <https://teaching.cornell.edu/teaching-resources/active-collaborative-learning/collaborative-learning>

Signature of Faculty Member

HOD